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ORIGINAL ARTICLE

Not all bull and bear markets are alike: insights from a five-state hidden semi-Markov model

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Abstract

This paper employs the hidden semi-Markov model and a novel model selection procedure to determine different regimes in the US stock market. The empirical results suggest that the US stock market is switching between five states that can be classified into three bull states and two bear states. The three bull states are categorized as a low-volatility bull market, a high-volatility bull market, and a stock market bubble. One of the bear states represents a regular bear market, while the other corresponds to either a stock market crash or a market correction. The paper demonstrates that the five-state model is consistent with a number of stylized facts and provides many valuable insights into the regime-switching dynamics of the US stock market and the risk-reward pattern of each regime. Besides, the paper demonstrates that the five-state model enables investors to make better asset allocation decisions. Specifically, in out-of-sample tests, the asset allocation strategy based on the five-state model achieves higher performance with lower risk than the strategy based on the two-state model and the buy-and-hold benchmark.

Keywords Semi-Markov models · Stock market regimes · Bull and bear markets · US stock market · Optimal asset allocation

JEL Classification C22 · G10 · G11

Introduction

For more than a century, investment professionals have regularly used the terms bull and bear markets to describe rising and falling price trends. Despite the widespread and prolonged usage of these terms, there is still no generally accepted formal definition of bull and bear markets. Only relatively recently, researchers have begun to propose various methods of dating bull and bear markets (see Chauvet and Potter

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2000; Maheu and McCurdy 2000; Hardy 2001; Pagan and Sossounov 2003; Lunde and Timmermann 2004; Gonzalez et al. 2005, among others). All these methods can be divided into two main types: methods that are based on using regime-switching Markov models (Chauvet and Potter 2000; Maheu and McCurdy 2000; Hardy 2001), and methods that are based on a set of rules (examples are Pagan and Sossounov 2003; Lunde and Timmermann 2004; Gonzalez et al. 2005). Both methods are of about the same popularity among researchers.

The methods based on a set of rules are motivated by the idea that a bull (bear) market denotes a period of generally increasing (decreasing) prices. There are basically only two distinct sets of rules. Specifically, Lunde and Timmermann (2004) suggest filter rules that track the percentage change between local tops and bottoms in the prices. For example, the stock market price should increase by more than 20% from the previous local bottom for the period to be deemed a distinct bull market. In contrast, Pagan and Sossounov (2003) and Gonzalez et al. (2005) insist that for the period to be considered a bull (bear) market, the stock market price should rise (fall) over a substantial period of time. These researchers adopt, with slight modifications, the complex set of rules used to identify turning points in the business cycle (Bry and Boschan 1971).

The methods based on using regime-switching Markov models are motivated by the observation that bull markets are characterized by low volatility and positive returns, whereas bear markets are associated with high volatility and negative returns. In contrast to the rule-based methods that analyze the price dynamics, the regime-switching methods examine the probability density function of the return distribution using a latent transition structure defined by a regime-switching Markov process. These methods classify return observations into several homogenous regimes characterized by different mean returns and volatilities. In the most simple case with two regimes, the Markov model assumes that the return process switches randomly between a low-volatility, high-return regime and a high-volatility, low-return regime.

The parameters of a regime-switching Markov model are typically estimated using the method of maximum likelihood implemented using various numerical optimization techniques. Quite frequently, the likelihood function is not well behaved and can have more than one peak. Besides, the algorithm requires that one specifies the starting values for all model parameters. Since the true model parameters are unknown in virtually all cases, the starting values are just an initial guess. If the initial guess is poor, the algorithm finds a local maximum instead of the global one. The estimated parameters of a regime-switching Markov model are also quite sensitive to the choice of a historical period. That is, small variations in a data sample's starting and ending dates can result in substantial variations in the estimated parameters of a Markov process.

In view of the aforesaid, dating of bull and bear markets using a regime-switching Markov model is often more art than science. By contrast, the methods based on a set of rules are fairly robust. Yet, in terms of describing the dynamics of a financial market, a great disadvantage of the rule-based methods is that they do not allow any complexity. Specifically, bull and bear markets detected by the rule-based methods are just some prolonged periods where the mean market return is either positive or negative. In reality, financial markets may exhibit very complex dynamics. Therefore, despite parameter-estimation problems, a great advantage of the regime-switching Markov models is that they allow a high degree of market dynamics complexity. The estimation of a Markov model with many regimes can provide valuable insights into the behavior of a financial market.

Since the initial application of a two-state regime-switching Markov model to identify bull and bear markets (Chauvet and Potter 2000; Maheu and McCurdy 2000; Hardy 2001), several studies extended the number of market states. For example, Dias et al. (2015) and Liu and Wang (2017) employ a three-state regime-switching model. Maheu et al. (2012) and Jiang and Fang (2015) operate with a four-state regime-switching model. Finally, De Angelis and Paas (2013) estimate a seven-state regime-switching model. Existing studies either specify a-priory the number of market states (examples are Maheu and McCurdy 2000; Maheu et al. 2012; Liu and Wang 2017) or select the number of states using the standard model selection criteria (Dias et al. 2015; De Angelis and Paas 2013).

All existing studies, except the study by Liu and Wang (2017), employ the Hidden Markov Model (HMM), where the sequence of market returns defines a Markov chain that satisfies the Markov property. A serious limitation of the HMM is that the state duration time has to follow the geometric distribution. As a consequence, in the HMM, there is no duration dependence. This means that the probability that a bull or bear market terminates does not depend on the market age. This feature contradicts the stylized facts about the duration dependence in bull and bear markets. Specifically, many studies (see Cochran and Defina 1995; Ohn et al. 2004; Lunde and Timmermann 2004; Harman and Zuehlke 2007; Zhou and Rigdon 2011; Zakamulin 2022) document the duration dependence in bull and bear markets. The majority of these studies find positive duration dependence. Specifically, they find that the higher the age of a bull (bear) market, the higher the probability that it terminates.

The hidden semi-Markov model (HSMM) generalizes the HMM by allowing the state duration time to follow any probability distributions. Therefore, the HSMM is better suited to capture the dynamics of a financial market. However, econometric applications of HSMMs to financial time series data remain scarce. To the authors' best knowledge, only the study by Liu and Wang (2017) employs the HSMM to detect various regimes in a stock market. In particular, Liu and Wang (2017) analyze the dynamics of the Chinese stock market by assuming that the market switches between three regimes: bull market, bear market, and sideways trending market.

The contributions of this paper are twofold. Our first contribution is to apply the HSMM to determine different regimes in the US stock market. In contrast to the study by Liu and Wang (2017), we select the number of market regimes using a novel model selection procedure. Our results suggest that the 5-state HSMM best describes the US stock market dynamics. The five states can be divided into three bull states and two bear states. The three bull states can be classified as a low-volatility bull market, a high-volatility bull market, and a stock market bubble. One of the two bear states represents a regular bear market. The other bear state can be categorized either as a stock market crash or a market correction.

The 5-state HSMM provides a number of valuable insights into the dynamics of the US stock market. For example, the model reveals that neither a low-volatility bull market nor a high-volatility bull market ends with a market crash, but a stock market bubble always ends up bursting. A bear market transits into a market crash with a probability of 56%. A market crash or correction is a short-lived phenomenon that lasts, on average, 3 months. This state of the market is typically followed by a period of market rally termed a "V-shaped recovery". Therefore, the 5-state HSMM justifies the conventional wisdom that says "buy the dips". Besides, the 5-state HSMM sheds additional light on the causes of the October 1987 stock market crash and the understanding of the Dot-com bubble of the 1990s. Last but not least, the 5-state HSMM exposes weaknesses of the popular volatility-based and trend-following strategies.

Our second contribution is to demonstrate that the 5-state HSMM enables academics and investment professionals not only to understand better the dynamics of the stock market but also to make better asset allocation decisions. Specifically, we consider the capital allocation between the stock market and the risk-free asset where the decision on the optimal stock investment depends on the risk-reward pattern of the prevailing market regime. Through an empirical study, we examine the profitability and riskiness of the strategy based on the 5-state HSMM and show that this strategy is superior not only to the buy-and-hold strategy but also to the strategy based on the conventional 2-state HSMM. In particular, we find that the performance and riskiness of the regime-based strategy based on the 2-state HSMM are virtually the same as those of the buy-and-hold strategy. In stark contrast, the 5-state HSMM strategy achieves higher performance with lower risk compared to the buy-and-hold strategy. In particular, the HSMM-5 strategy allows the investor to reduce the risk by 30–35% and increase the Sharpe ratio by 35–50%.

The rest of the paper is organized as follows. Section Data and descriptive statistics presents the data and descriptive statistics. Section Methodology describes the methodology. In particular, this section presents the HSMM and our novel model selection procedure. Section Empirical Results reports the empirical results. Section Discussion presents the discussion and elaborates on a number of valuable implications. Section Regime-Based Asset Allocation examines the profitability and riskiness of the regime-based asset allocation strategies. Finally, Section Summary and Conclusions summarizes and concludes the paper.

Data and descriptive statistics

Our main study uses the monthly capital gain returns on the Standard and Poor's (S&P) 500 index from January 1957 to December 2020. The data are provided by the Center for Research in Security Prices (CRSP). The S&P 500 index, introduced in 1957, is a capitalization-weighted index that tracks the value of 500 large-cap companies in the US. This index is perhaps the most commonly followed stock market index, and many consider it the best representation of the US stock market.

Table 1 shows the descriptive statistics of the monthly capital gain returns on the S&P 500 stock market index. In particular, this table reports the mean, standard deviation, skewness, and (excess) kurtosis of the returns on the S&P 500 index. In annualized terms, the mean return is 7.91%, and the standard deviation is 14.64%.

Table 1 Descriptive statistics of the monthly capital gain returns on the S&P 500 stock market index

Mean return	7.91
Standard deviation	14.64
Skewness	− 0.43
Kurtosis	1.72

The mean return and standard deviation are annualized and reported in percentages. *Kurtosis* is the excess kurtosis which is defined as the kurtosis of a normal distribution minus 3

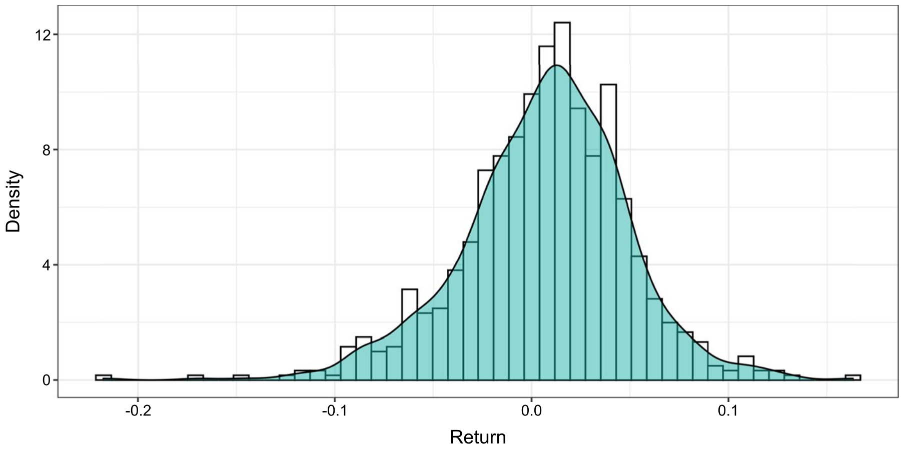

Fig. 1 Empirical kernel density of the monthly returns on the S&P 500 stock market index

The skewness is negative, and the excess kurtosis is positive. The signs of these two statistics imply that the returns on the S&P 500 index are leptokurtic, and the left tail of the return distribution is fatter than the right tail. Figure 1 plots the empirical kernel density of the monthly returns on the S&P 500 stock market index.

Methodology

Hidden semi-Markov model

Traditional modeling of market returns, including the random walk model, assumes that the returns are independently and normally distributed. Formally, if P_t and P_{t+1} are the market index prices at times t and $t+1$ respectively, then the (capital gain) return is

$$x_t = \frac{P_{t+1} - P_t}{P_t} \sim \mathcal{N}(\mu, \sigma^2),$$

where μ and σ are the mean return and standard deviation of returns, respectively. However, empirical studies indicate that this model fails to reproduce several well-documented stylized facts: fat tails, short-term momentum, volatility clustering, etc.

A regime-switching Markov model allows the return process to switch randomly between J regimes with different values for μ and σ. Specifically, in state $j \in [1, \dots, J]$ the return is given by

$$x_t | S_t = j \sim \mathcal{N}(\mu_j, \sigma_j^2),$$

where S_t denotes the market state at time t, and μ_j and σ_j are the mean return and standard deviation of returns, respectively, in state j. In the HMM, a sequence of discrete random variables $\{S_t\}$ defines a Markov chain that satisfies the Markov property

$$Prob(S_{t+1} | S_t, S_{t-1}, \dots, S_1) = Prob(S_{t+1} | S_t).$$

In words, the conditional probability of the next state depends only upon the present state, not on the sequence of states that preceded it. The conditional probabilities $p_{ij} = Prob(S_{t+1} = j | S_t = i)$ are called the transition probabilities. Additionally, in the HMM, the state process $\{S_t\}$ is specified by the initial probabilities $\pi_j = Prob(S_1 = j)$ with $\sum_j \pi_j = 1$. Timmermann (2000) demonstrates that even a 2-state HMM for market returns can reproduce most of the well-documented stylized facts.

A serious limitation of the HMM is that the state duration time[1] has to follow the geometric distribution. The HSMM generalizes the HMM by allowing the state duration time to follow any probability distributions. Unlike the HMM, one assumes that the transition probability from one state to the same state in the HSMM is zero, $p_{ii} = 0$. Consequently, the diagonal elements in the transition probability matrix are zeros:

$$\mathbf{P} = \begin{pmatrix} 0 & p_{12} & \cdots & p_{1J} \\ p_{21} & 0 & \cdots & p_{2J} \\ \vdots & \vdots & \ddots & \vdots \\ p_{J1} & p_{J2} & \cdots & 0 \end{pmatrix}.$$

Contrary to the HMM, the HSMM does not have the Markov property at each time t. The Markov property is satisfied only when the process changes the state. In particular, when a process enters a state i, it determines the next state j according to the transition probability p_{ij}. However, after j has been selected, but before making the transition, the process holds in state i for a time τ_i. In the HSMM, the distribution of the state duration time τ_i is governed explicitly by a probability density function:

$$d_i(m) = Prob(\tau_i = m), \quad m = 1, 2, 3, \dots,$$

where m denotes the duration time. Nystrup et al. (2015) demonstrate that the HSMM reproduces the stylized facts about the market returns better than the HMM.

[1] The duration time is also known as the dwell time, occupancy time, holding time, or sojourn time.

Our estimation of the HSMM is based on the R package `hsmm` (Bulla et al. 2010). This package implements only a few probability distributions for the state duration times. Specifically, besides the geometric distribution, the duration time can be modeled by either the logarithmic, negative binomial, or Poisson distribution. Our choice is the Poisson probability distribution. The reason for this choice is twofold. First, the Poisson probability distribution implies the required positive duration dependence. Second, this distribution has only one parameter. Therefore, this allows one to avoid overfitting and produce stable estimation results. The probability density function of the Poisson distribution is given by

$$d(m, \lambda) = Prob(\tau = m) = \frac{\lambda^m e^{-\lambda}}{m!}, \quad m = 1, 2, 3, \dots,$$

where m is the state duration time. The expected duration time is $E[\tau] = \lambda$. That is, the parameter λ defines the average state duration.

Model selection procedure

Model selection involves determining the optimal number of market states J that best describes the dynamics of the market. We start this procedure by varying the number of market states $J \in [2, J_{\max}]$ where $J_{\max}$ is the arbitrarily chosen maximum number of market states. For each value of J, the parameters of the HSMM model are estimated using the maximum likelihood method. The goodness of fit of the HSMM model with J market states is measured by the log-likelihood function denoted by $LL(J)$.

The problem is that the log-likelihood function increases with the increased number of market states. This is because the greater the number of market states, the greater the number of model parameters, and the better the goodness of fit. Therefore, the log-likelihood value cannot be used as a model selection criterion. Typically, to determine the optimal number of market states in either the HMM or HSMM, one uses the model selection criteria to find the best tradeoff between the goodness of fit and the number of model parameters. For example, the Akaike Information Criterion (AIC) for the model with J states is computed as follows

$$AIC(J) = 2K(J) - 2LL(J),$$

where $K(J)$ is the number of parameters in the HSMM model with J states. The AIC rewards the goodness of fit but penalizes the number of estimated parameters. This criterion is rigorously founded on information theory and tries to capture in the simplest possible way the intuition that new parameters must be worthwhile in terms of likelihood improvement from the principle of parsimony.[2] The AIC is very popular

[2] We remind the reader that the best model is the model with the smallest value of AIC.

because it greatly simplifies the model selection procedure. However, the AIC says nothing about the quality of the selected model.

We argue that finding the most parsimonious model makes little sense when the core idea is finding a model that most correctly describes the dynamics of a financial market.[3] Instead of using the AIC or similar model selection criteria, we suggest a novel approach to select the optimal number of states in the HSMM model. We acknowledge that our approach is not rigorously founded and, therefore, can be criticized. However, this approach produces a realistic model that is highly consistent with both scientific and anecdotal evidence as well as popular and expert opinions about the stock market dynamics.

Now we turn to the description of our approach to model selection. The outcome of the estimation of the HSMM model with J market states is not only the values of the estimated parameters but also the decoded sequence of market states $\{S_t\}$. If in state j, $1 \leq j \leq J$, the estimated $\mu_j > 0$ ($\mu_j < 0$) then the logical interpretation is that state j is a bull (bear) state. Assume, for example, that the algorithm detects that the market enters the state j at time t and exits this state at time $t + \tau + 1$. Specifically, we assume that

$$S_{t-1} \neq j, S_t = j, \dots, S_{t+\tau} = j, S_{t+\tau+1} \neq j.$$

That is, over the period $[t, t + \tau]$ the decoded market state is j. Therefore, we naturally expect that over this period, the sign of the mean return must equal the sign of the estimated mean return in state j. For instance, if $\mu_j > 0$ then the mean return over $[t, t + \tau]$ must be positive. Put differently, if the period $[t, t + \tau]$ is labeled as a bull state, then the market price must increase, not decrease, over this period. However, due to randomness, the algorithm finds the most probable state sequence, and there is no guarantee whatsoever that over $[t, t + \tau]$ the mean return is positive. That is, the mean return over the period $[t, t + \tau]$ may happen to be negative even though this period is decoded as a bull state. If this is the case, we say that the market state over $[t, t + \tau]$ is decoded incorrectly. The lesser the number of incorrectly decoded market states, the better the HSMM model's quality. Therefore, we argue that the most natural model selection criterion should be based on the number of correctly decoded stock market states.

Formally, our proposed novel model selection criterion is the percentage of correctly decoded states of the stock market computed as

$$\frac{C(J)}{N(J) + 1} \times 100,$$

where $N(J)$ is the number of times the J-state HSMM changes state[4] and $C(J)$ is the number of correctly decoded states out of totally observed $N(J) + 1$ distinct states.

[3] A somewhat similar critique of using the standard model selection criteria to determine the number of states in the HMM is presented by Pohle et al. (2017). These authors emphasize, among other things, that in selecting the number of states, one should consider the realism of the fitted candidate models assessed using expert knowledge.

[4] In other words, $N(J)$ is the number of transitions between the states.

For the sake of illustration of the idea, suppose that in the HSMM with $J = 2$ states, the decoded sequence of states is as follows:

Time, t	1	2	3	4	5	6	7	8	9	10	11	12
State, S_t	1	1	2	2	2	2	1	1	1	2	2	2

That is, over the data sample that contains 12 monthly observations, the initial decoded state is 1, and the HSMM changes state 3 times. Specifically, the market enters state 1 at time 1, holds in state 1 for 2 months, and then transits to state 2 at time 3. It stays in state 2 for 4 months and then moves to state 1 at time 7. The market remains in state 1 for 3 months before transitioning to state 2 at time 10. Over the last three months, the market remains in state 2.

Introducing the variable $k \in [1, N(J) + 1]$, the decoded sequence of states can be described by the times of entering a distinct state $t(k)$; the holding times in each distinct state $\tau(k)$; and the indices of decoded states $j(k) \in [1, J]$. In our example $k = \{1, 2, 3, 4\}$, $t(k) = \{1, 3, 7, 10\}$, $\tau(k) = \{2, 4, 3, 3\}$, and $j(k) = \{1, 2, 1, 2\}$. Consequently, the number of correctly decoded states is computed as

$$C(J) = \sum_{k=1}^{N(J)+1} I\left(\operatorname{sgn}\left(\frac{1}{\tau(k)} \sum_{t=t(k)}^{t(k)+\tau(k)-1} x_t \right) = \operatorname{sgn}\left(\mu_{j(k)}\right) \right),$$

where x_t is the market return over $[t, t + 1]$, $\operatorname{sgn}(\cdot)$ is the mathematical sign function, and $I(\cdot)$ denotes the indicator function that takes the value of one if the condition is satisfied and zero otherwise. It is worth repeating that the idea is to count the number of times the sign of the mean return in each decoded distinct state equals the sign of the estimated mean return over this state.[5]

In concluding this section, let us illustrate the computation of the percentage of correctly decoded states in the example above. Suppose the estimated mean returns over the two states are $\mu_1 < 0$ and $\mu_2 > 0$. Thus, we label state 1 as a bear market whereas state 2 as a bull market. Suppose further that

$$\frac{x_1 + x_2}{2} > 0, \quad \frac{x_3 + x_4 + x_5 + x_6}{4} > 0, \quad \frac{x_7 + x_8 + x_9}{3} < 0, \text{ and } \frac{x_{10} + x_{11} + x_{12}}{3} > 0.$$

Note that the first two months in the sample are decoded as a bear state. However, the return over these months is positive, which means that this bear state is decoded incorrectly. The other three distinct states are decoded correctly. Therefore, in our example, the percentage of correctly decoded states is $3/4 \times 100 = 75\%$.

[5] The estimated mean return in state j is computed as the mean return over all months in the data sample decoded as being state j.

Table 2 For each *J*-state HSMM, this table reports the log-likelihood value (LL), the AIC value, and the percentage of correctly decoded states

Number of states	LL	AIC	Correctly decoded states
2	1374.99	– 2729.97	90.91
3	1405.15	– 2774.30	83.33
4	1407.44	– 2758.88	82.81
5	1419.14	– 2758.29	93.22
6	1425.73	– 2743.46	84.44
7	1429.31	– 2718.62	81.63

Table 3 Estimated parameters of the 5-state HSMM for the S&P 500 index returns

	Number of occurrences	Percentage of time	Mean return μ	Standard deviation σ	Average state duration λ
State 1	8	19.15	16.06	6.65	16.73
State 2	14	33.03	– 3.06	12.83	16.51
State 3	8	12.21	26.56	12.70	10.12
State 4	16	7.07	– 30.23	25.31	2.98
State 5	12	28.53	19.74	13.77	17.05

The mean returns and standard deviations are annualized and reported in percentages. The average state duration is measured in months. *Number of occurrences* denotes the number of times a particular market state is observed. *Percentage of time* denotes the percentage of time the market holds in a specific state

Empirical results

We estimate the HSMM for $J \in [2, 7]$ latent states. For each J-state HSMM, Table 2 reports the log-likelihood value, the AIC value, and the percentage of correctly decoded states. As the number of states increases, the goodness of fit, as measured by the log-likelihood, increases. However, the number of model parameters also increases with the number of states. According to the AIC criterion, the HSMM with 3 states provides the best tradeoff between the goodness of fit to the data and the number of estimated parameters. However, the quality of the 3-state HSMM, as measured by the percentage of correctly decoded states, is far from the best among the candidate models. Specifically, according to the percentage of the correctly decoded state's criterion, the 3-state HSMM is the 4th best among the candidate models. The HSMM with 5 states has the highest percentage of correctly decoded states. In particular, whereas the 3-state HSMM correctly identifies 83% distinct states, the 5-state HSMM correctly identifies 93% distinct states. Therefore, our choice is the 5-state HSMM.

Table 3 presents the estimated parameters of the 5-state HSMM for the monthly returns on the S&P 500 index. Namely, this table reports the mean return, the standard deviation of returns, and the average state duration for each state. In addition, this table shows the number of times a particular market state is observed and the

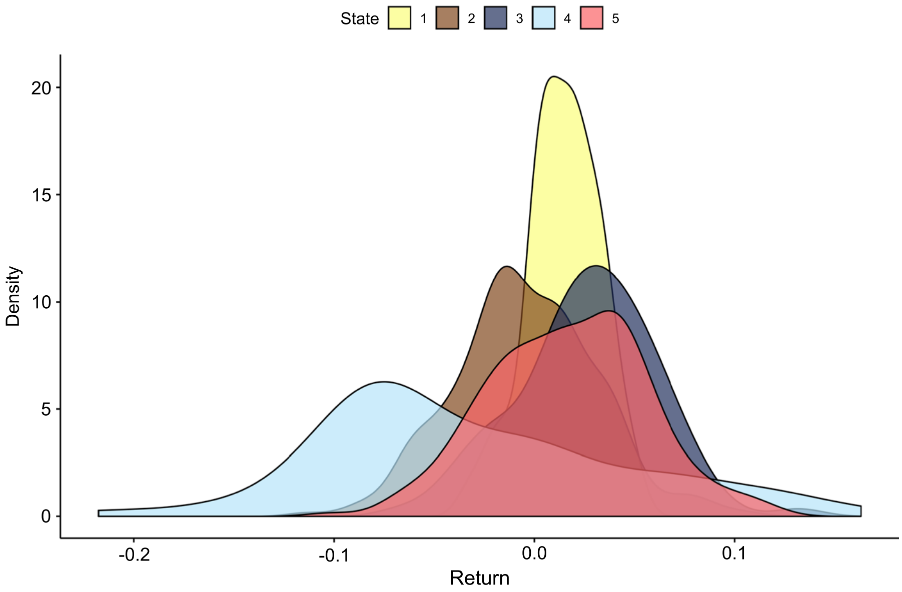

Fig. 2 Empirical kernel state densities of the monthly returns on the S&P 500 stock market index in the 5-state HSMM

Table 4 Estimated transition probability matrix in the 5-state HSMM for the S&P 500 index

	State 1	State 2	State 3	State 4	State 5
State 1	0.00	0.84	0.16	0.00	0.00
State 2	0.29	0.00	0.15	0.56	0.00
State 3	0.00	0.00	0.00	1.00	0.00
State 4	0.00	0.00	0.36	0.00	0.64
State 5	0.35	0.65	0.00	0.00	0.00

percentage of time the market holds in a specific state. For example, this table reveals that the market was 8 times in state 1; the average duration of this state amounts to 16.7 months; and the market holds about 19% of the time in this state. For each market state, Fig. 2 plots the empirical kernel state densities of the monthly returns on the S&P 500 stock market index. The kernel density for each state is plotted in a specific color in this plot.

Based on the results reported in Table 3, the following observations deserve mentioning. Our first observation is that in two states (2 and 4) the mean market return is negative, whereas in the other three states (1, 3, and 5) the mean market return is positive. Therefore, we conclude that the 5-state HSMM has two bear market states and three bull market states. About 42.6% of the time, the market happens to be in a bear state; the other 58.4% of the time, the market is in a bull state.

Our second observation is related to the value of mean returns in bull and bear states. Among three bull market states, States 1 and 5 have about the same value of

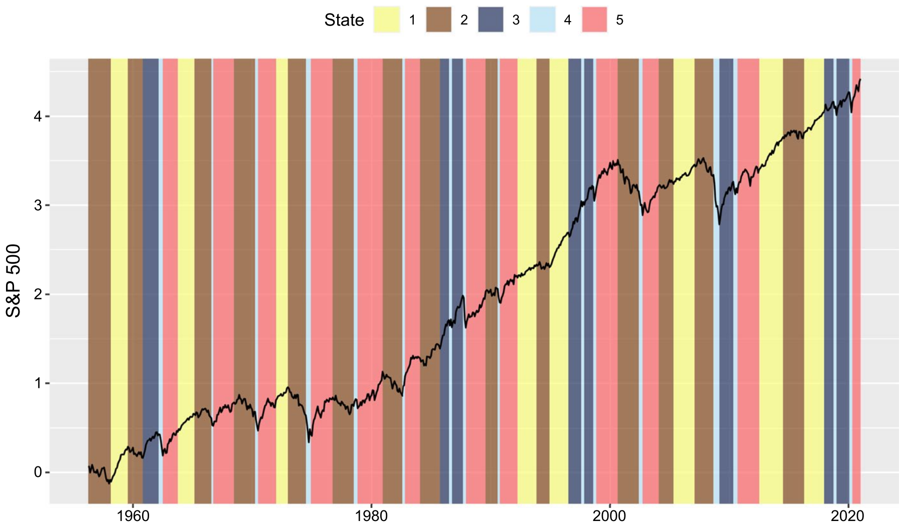

Fig. 3 Cumulative return on the S&P 500 index from 1957 to 2020 and the market states in the 5-state HSMM

Table 5 Proposed classification of market states in the 5-state HSMM for the S&P 500 index

	Classification
State 1	Low-volatility bull market
State 2	Bear market
State 3	Stock market bubble
State 4	Stock market crash or market correction
State 5	High-volatility bull market

mean returns (16 and 19.7%), whereas in State 3, the mean market return is notably higher (26.5%). In two bear market states, State 2 has a relatively small negative mean return (− 3%), whereas State 4 is characterized by an extremely low negative mean return (− 30%).

Our third observation is that, roughly, the market has three levels of volatility (standard deviation of returns). Specifically, State 1 is a state with low volatility (6.6%). States 2, 3, and 5 are high-volatility states with about the same volatility levels (12.8, 12.7, and 13.8%). Finally, State 4 is characterized by extremely high volatility (25.3%).

Our fourth observation concerns the average state durations. States 1, 2, and 5 have approximately the same average state durations (16.7, 16.5, and 17.1 months). The average duration of State 3 is almost twice as short (10.1 months) as that of States 1, 2, and 5. Finally, State 4 is very short-lived, being on average 3 months.

Table 4 reports the estimated transition probability matrix in the 5-state HSMM for the S&P 500 index. For example, this table reveals that, after being in State 1, the market transits to either State 2 with a probability of 84% or State 3 with a probability of 16%. Figure 3 plots the cumulative return on the S&P 500 index and the market states in the 5-state HSMM. In this plot, each state of the market is highlighted by a specific color. The color palette in this figure is the same as that in Fig. 2. Finally, Table 7 in the Appendix presents the dating of decoded states in the 5-state HSMM for the S&P 500 index.

Motivated by the descriptive statistics of market states and the estimated transition probability matrix between the states, the suggested classification of market states is presented in Table 5. In particular, we propose to classify the three bull market states as follows. State 1 is a low-volatility bull market, State 5 is a high-volatility bull market, and State 3 is a stock market bubble. The proposed classification of the two bear market states is as follows. State 2 is a bear market, whereas State 4 is either a stock market crash or a market correction.

The proposed classification of States 1, 2, and 5 is rather obvious. Specifically, States 1 and 5 are bull market states with about the same mean returns, but the volatility in State 5 is approximately twice as high as that in State 1. Therefore, we classify States 1 and 5 as the low-volatility bull market and the high-volatility bull market, respectively. State 2 is a high-volatility, negative-return state that can be naturally classified as a bear market.

The justification of the suggested classification of States 3 and 4 deserves a more detailed explanation. First, we note that State 3 is a high-volatility bull market with a higher mean return than the mean return in the other two bull markets. Therefore, prices in this bull market increase more rapidly than in the other two bull markets. Second, the average duration of State 3 is almost twice as short as that in the other two bull markets. Third and finally, after being in State 3, the market always transits to State 4 (see Table 4), which is a short-lived market state where the prices decline significantly. Consequently, quite a straightforward interpretation of the transition between States 3 and 4 is as follows. State 3 is a market bubble where prices increase dramatically over a rather short period, and this increase is not supported by fundamentals. When the investors recognize that the market prices are significantly above intrinsic values, the bubble “bursts”, and prices crash. This market crash represents a short-lived and strong market correction captured by State 4. The subsequent section discusses various transitions between the market states in more detail and gives examples of historical episodes that illustrate the typical market dynamics.

Discussion

We start the discussion by noting that, in the formal bull-bear dating algorithms based on a set of rules, there is only one bull state and one bear state; a bull state always transits to a bear state and vice versa. In a 5-state HSMM, on the other hand, a bull state can transit to another bull state, and a bear state can transit to another bear state. As a result, very often, a bull (bear) market state identified by a dating algorithm consists of two different bull (bear) market states in the 5-state HSMM. That is, one type of a bull (bear) market follows after another type of a bull (bear) market. For illustration, Fig. 4 shows the market states of the S&P 500 index from

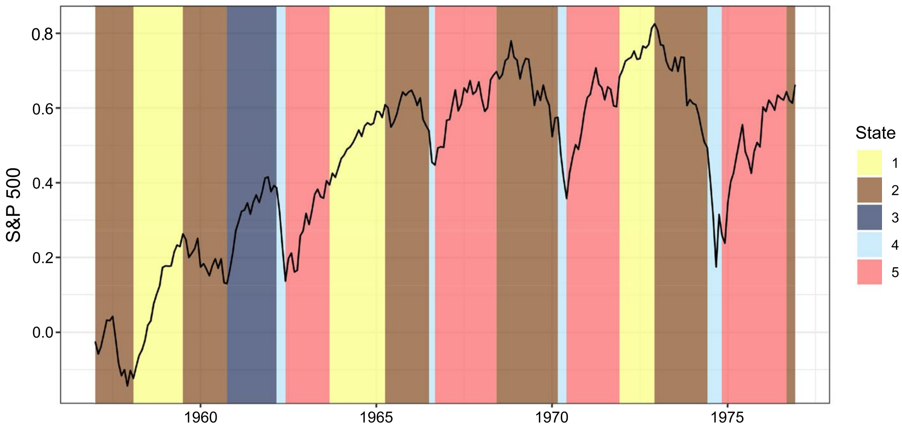

Fig. 4 Markets states over the period from 1957 to 1976

1957 to 1976. One can notice that this period encompasses two historical episodes where a high-volatility bull market transits to a low-volatility bull market. Specifically, the first historical episode covers the period from July 1962 to February 1965, and the second historical episode covers the period from July 1970 to December 1972. The transition probability matrix presented in Table 4 reports that a high-volatility bull market transits to a low-volatility bull market with a probability of 36%. In addition, a low-volatility bull market transits to a stock market bubble (another bull state) with a probability of 16%.

Moreover, the period from 1957 to 1976 encompasses three historical episodes where a bear market (State 2) is followed by a market crash (State 4), representing another kind of a bear market. That is, at the end of a bear market, prices often decline steeply, and this period of steep decline is captured by State 4 in the HSMM. As the transition probability matrix reveals, a bear market develops into a market crash with a probability of 56%. The investors' psychology can explain the occurrence of such a market crash. In particular, throughout a bear market, the prices decrease, and some investors begin to sell stocks. Subsequently, other investors join the crowd and start selling stocks as well. The overall sale drives down stock prices, and the resulting drop in prices drives pessimism among investors. The pessimism can develop into an extreme fear that causes a market crash due to a positive feedback effect (a.k.a. the "bandwagon" effect).

The transition probability matrix documents that State 4 follows only after State 2 (bear market) or State 3 (market bubble). Neither a low-volatility bull market nor a high-volatility bull market ends with a market crash. Although there are three types of bull markets in the 5-state HSMM, two types of bull markets never end with a market crash, but a stock market bubble always ends up bursting. In this regard, the results of the estimation of the 5-state HSMM shed additional light on the causes of the October 1987 stock market crash.

Specifically, one of the most often cited causes of this crash is "portfolio insurance", a popular hedging strategy at that time executed by computers (see, for

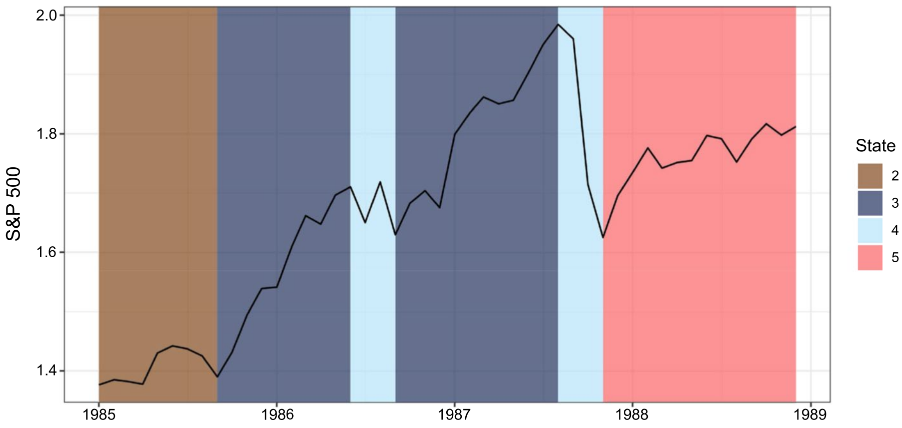

Fig. 5 Markets states over the period from 1985 to 1988

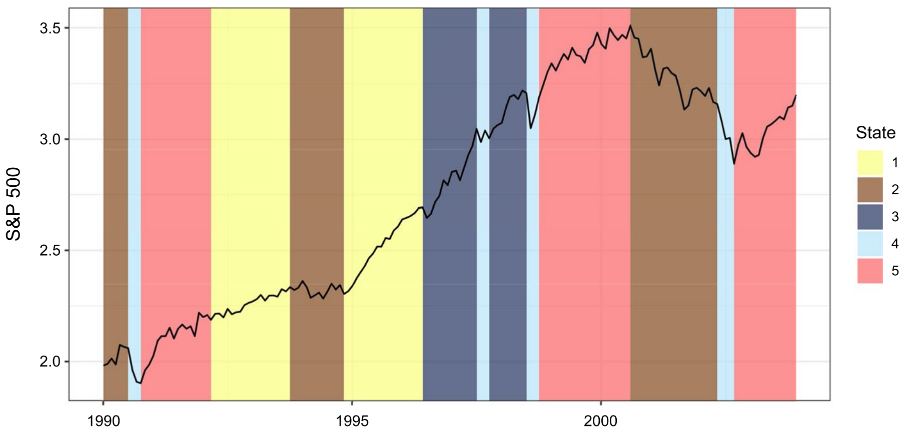

Fig. 6 Markets states over the period from 1990 to 2003

example, Shiller 1988; Carlson 2007). However, some investment professionals and academics also point to the fact that by October 1987, the market was overvalued and due for a major correction (Shiller 1988). Sornette et al. (1996), and subsequently, Phillips et al. (2015) developed econometric techniques for detecting financial bubbles and demonstrated the existence of a market bubble before the stock market crash in October 1987. The results of the estimation of the 5-state HSMM also reveal the presence of a market bubble over the period from October 1985 to June 1986 and subsequently over the period from October 1986 to August 1987 (see Fig. 5). Consequently, one can argue that the stock market crash in October 1987 was a necessary stock market correction after a bubble built up earlier.

The results of the 5-state HSMM's estimation also shed additional light on understanding the Internet (Dot-com) bubble of the 1990s, its rise, and burst. Since this bubble was caused by excessive speculation in Internet-related companies, the dating of the bubble is usually done by examining the dynamics of the Nasdaq Composite stock market index. Judging by the Nasdaq index's behavior, the bubble started in 1995 and ended with a market crash in March 2000. In this regard, an interesting question is whether there was a similar rise and burst in the S&P 500 index. The results obtained by Phillips et al. (2015) suggest the existence of a bubble in the S&P 500 index from December 1995 to October 2001. However, these researchers detect financial bubbles by studying the dynamics of the dividend-to-price ratio. By contrast, the HSMM uses returns to identify the state of the market.

Figure 6 shows the market states of the S&P 500 index from 1990 to 2003. The 5-state HSMM decodes the period from December 1994 to June 1996 as the low-volatility bull market. However, the mean market return over this period was rather high. Therefore, judging by the value of mean returns, a more appropriate classification of this period could be a "low volatility market bubble". The HSMM identifies that the bubble in the S&P 500 index lasted from July 1996 to July 1998, interrupted by a mini-crash caused by an economic crisis in Asia (August 1997 to October 1997). The bubble ended with a market crash from August 1998 to October 1998. This market crash was caused by the Russian default and subsequent global currency crisis. After this crisis, the S&P 500 index entered the high-volatility bull market that transited to a bear market in September 2000. This bear market ended with a market crash that lasted from June 2002 to September 2002. Consequently, the 5-state HSMM suggests that the S&P 500 index bubble ended with the market crash in 1998. That is, it ended much earlier than the bubble in the Nasdaq index.

The results of the 5-state HSMM's estimation, reported in Table 3, suggest that a market crash/correction occurs quite often. In particular, from 1957 to 2020, a market crash/correction (State 4) happened 16 times. During a crash/correction, the market prices drop significantly. However, a market crash/correction is a short-lived market phenomenon that varies, on average, 3 months. In this regard, one can argue that a market crash at the end of a bear market signals that the price decline will be over soon. The transition probability matrix documents that State 4 transits to a high-volatility bull market with a probability of 65% or to a market bubble with a probability of 35%. In either case, conventional wisdom says that a market crash/correction is followed by a period of a market rally. Therefore, investment professionals refer to this transition as a "V-shaped recovery".

A visual inspection of the dynamics of the S&P 500 index (see Figs. 3, 4, 5, 6) supports the observation that after a market crash the index rebounds very strongly. To validate the conventional wisdom that says "buy the dips," we compute the mean return on the index over the first 6 months of each state that follows State 4. In annualized terms, this mean market return amounts to 36%. Such a high estimate confirms the presence of a market rally over the first 6 months after a market crash/correction.

Again, the investors' psychology can explain the occurrence of a market rally after a market crash/correction. In particular, the researchers documented that investors are prone to overreaction (see De Bondt and Thaler 1985; Daniel et al. 1998, and many others). As originally formulated by De Bondt and Thaler (1985), if investors systematically overreact, the stock price dynamics must satisfy the following two properties. First, extreme movements in stock prices must be followed by subsequent price movements in the opposite direction. Second, the more extreme the initial price movement, the greater the following adjustment.

The pattern of a “V-shaped recovery” is consistent with the overreaction hypothesis. For example, a plausible story behind a market rally right after a market correction is as follows. A correction is supposed to bring the market back to normal levels. However, during a correction, the investors tend to overreact, and market prices drop much below their fundamental values. Eventually, investors realize that the market is highly undervalued, and the stocks become a bargain. They see a buying opportunity and start bidding prices up. Their sentiment undergoes a sharp change from extremely bearish to extremely bullish. This change in sentiment marks the end of the overreaction phase and the start of the phase where stock prices recover quickly to the levels supported by fundamentals.

In concluding this section, we would like to elaborate on the implications of the 5-state HSMM for the success of some active strategies that aim to enhance returns and reduce losses. First, we consider the volatility-based strategies. These strategies are motivated by the existence of an inverse relationship between returns and volatility. In particular, when volatility is high, the returns are low and vice versa (French et al. 1987). This empirical evidence seems to agree with the conventional wisdom that a bull market is a high-return low-volatility state, whereas a bear market is a low-return high-volatility state. Therefore, volatility-based strategies monitor the volatility level and reduce (enhance) exposure to stocks when volatility is high (low). Empirical studies often report that volatility-based strategies outperform the passive buy-and-hold strategies (see, for example, Collie et al. 2011; Butler and Philbrick 2012; Albeverio et al. 2013; Zakamulin 2014; Perchet et al. 2015; Moreira and Muir 2017).

The 5-state HSMM reveals one potentially serious issue with volatility-based strategies. Specifically, whereas the volatility is indeed high in bear market states, the volatility is not low in all bull markets. In fact, in the majority of bull markets, the volatility is also high. To be specific, we remind the reader that the market is in a bull state 58.4% of the time. The low-volatility bull market occurs 19.5% of time (see Table 3). Consequently, the volatility is high over the remaining 38.9% of the bull market time. Therefore, if the market is in a bull state, then the volatility is low only about 33% of the time. The other 67% of the time, the volatility is the same as in the bear market (except the periods of stock market crashes and corrections). A prominent example in this regard is the period from 1996 to 2000, over which both the market return and volatility were high. Historical simulations show that a volatility-based strategy significantly reduces the exposure to stocks over this period and, therefore, misses the high returns.

Second, we consider trend-following strategies. These strategies are based on the common belief that stock prices tend to move in trends. To detect trends, investors typically use moving averages of prices. The most popular trend-following rule is the 50- and 200-day crossover rule. This rule generates a Sell (Buy) signal when the 50-day simple moving average is below (above) the 200-day simple moving average. The problem is that any trend-following rule identifies a turning point in a trend with some delay. Therefore, for a trend-following strategy to generate profitable

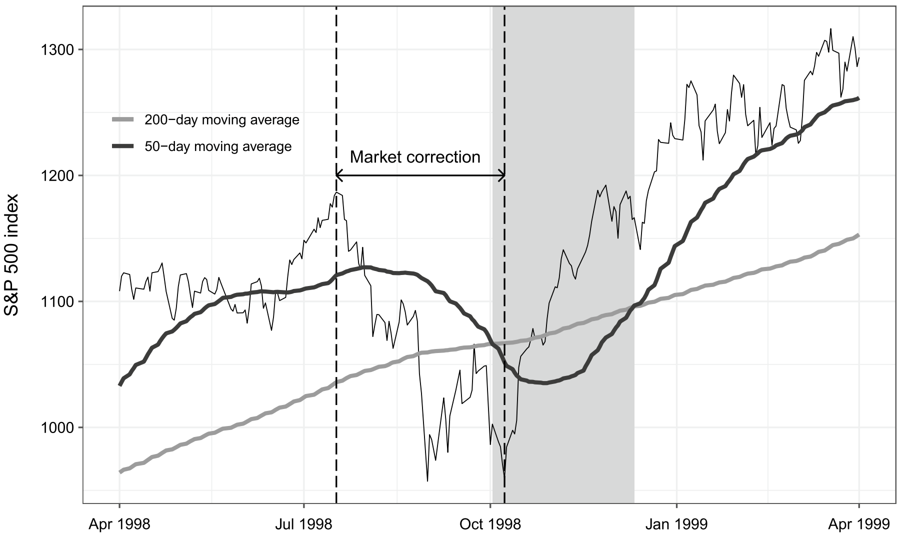

Fig. 7 The daily values of the S&P 500 index over the period from April 1998 to April 1999 and the values of 50- and 200-day simple moving averages. The shaded area highlights the period where the crossover rule generates a Sell signal

trading signals, the trend duration must be notably longer than the delay time. If this condition is not satisfied, the trend-following strategy does the opposite of what it is supposed to do. An example that supports this point is the performance of the 50- and 200-day crossover rule throughout short-lived market corrections. In particular, this rule works extremely poorly over the periods where a market bubble transits to a market correction that ends with a market rally.

For illustration, Fig. 7 plots the daily prices of the S&P 500 index from April 1998 to April 1999 and the values of 50- and 200-day simple moving averages. This period encompasses a short market correction triggered by the 1998 Russian default. The 5-state HSMM decodes the period from August 1998 to October 1998 as a market correction. More precisely, the market bubble burst on the 17 of July, and the subsequent market correction lasted til the 8 of October, less than 3 months. The problem is that the crossover rule generated a Sell signal only on the 2 of October 1998, close to the end of the market correction. The subsequent Buy signal was generated on the 11 of December 1998. Therefore, virtually the whole Sell period overlapped with the market rally. As a result, the popular crossover rule was not able to protect the trend following investors from losses suffered during the market correction. Besides, these investors missed the subsequent strong market rally.

Regime-based asset allocation

In this section, we consider the capital allocation between the stock market and the risk-free asset where the decision on the optimal stock investment depends on the risk-reward pattern of the prevailing market regime. The theoretical framework for the study in this section is as follows. We consider an investor who decides how to allocate optimally between the stock market and the risk-free asset. The investor's time t capital allocation consists in investing fraction y_t in the stocks and the rest, $1 - y_t$, in the risk-free asset. Thus, the time t return to the investor's portfolio equals

$$r_{p,t} = y_t r_t + (1 - y_t) r_{ft},$$

where r_t and r_{ft} are the time t returns on the stocks[6] and the risk-free asset, respectively.

We assume that our investor has mean-variance preferences. Therefore, the investor's goal is to select the fraction y_t to maximize the mean-variance utility of the portfolio. Specifically, the one-period utility maximization problem is given by

$$\max_{y_t} \quad U(r_{p,t}) = E[r_{p,t}] - \frac{\gamma}{2} Var[r_{p,t}],$$

where $E[r_{p,t}]$ is the expected portfolio return, $Var[r_{p,t}]$ is the variance of portfolio, and γ is the investor's level of risk aversion. The solution to the optimal fraction y_t is given by (see, for example, Bodie et al. (2013, Chap. 6))

$$y_t = \frac{\mu_t - r_{ft}}{\gamma \sigma_t^2},$$

where μ_t and σ_t denote the time t mean return and standard deviation of the stock market returns.

It is essential to note that we assume that the parameters of the stock market returns μ_t and σ_t are time-varying and known to the investor. Our investor solves not a single-period utility maximization problem but a multi-period utility maximization problem. However, in the absence of serial dependence in stock returns, the investor behaves myopically, which means that the multi-period capital allocation problem reduces to a sequence of single-period problems, see Zakamulin (2016).

Note that the time t optimal fraction y_t is directly proportional to the time t risk premium (as measured by $\mu_t - r_{ft}$) offered by the risky stocks and inversely proportional to the level of risk aversion γ and the time t level of risk (as measured by σ_t^2). Motivated by the fact that the stock volatility is predictable, Zakamulin (2016) uses this theoretical framework to justify volatility-based strategies. The main shortcoming of a volatility-based strategy is that it leaves out the mean stock return. The mean stock return is not included because there is a common understanding among

[6] Note that r_t denotes the time t total return on the stocks that consists of the capital gain return x_t and the dividend yield. In this section, variables μ and σ denote the mean total return and the standard deviation of the total returns.

academics and investment professionals that the mean return is not predictable. Regime-switching models allow the investor to predict (to some extent) not only the stock market volatility but also the stock market mean return. Consequently, regime-based asset allocation fits the theoretical framework better than volatility-based asset allocation.

There are mainly two alternative approaches to regime-based asset allocation. One straightforward approach is to use the Viterbi algorithm and find the most likely state of the market at time t, which denotes the end of the in-sample period. Since the market states are persistent, we assume that the market will be in the same state at the next time $t+1$. In this approach, the optimal fraction to be invested in the stocks for the next period is given by

$$y_{t+1} = \max\left(\frac{\mu_{j(t)} - r_{ft}}{\gamma\sigma^2_{j(t)}}, 0\right),$$

where $j(t)$ is the decoded state of the market at time t and $\mu_{j(t)}$ and $\sigma_{j(t)}$ are the estimated mean return and standard deviation of the market returns in state $j(t)$. Note that we do not allow short sales. Therefore, the optimal fraction cannot be negative.

The other more sensible approach is to use the probabilities that the market is in state j at time t

$$Prob(S_t = j), \quad j = 1, 2, \dots, J.$$

These probabilities are called "smoothing probabilities". We justify this approach by the following example. Assume a 2-state HSMM. Suppose that the smoothing probabilities of each possible state are $Prob(S_t = 1) = 49\%$ and $Prob(S_t = 2) = 51\%$. The most likely market state is 2. However, in this situation, the market can occupy either state with virtually equal probabilities. Due to high uncertainty in the market state, it is hazardous to choose the optimal fraction assuming that the market occupies the state with the highest probability. It is better to use the average optimal fraction across all possible states weighted by their smoothing probabilities

$$y_{t+1} = \max\left(\sum_{j=1}^{J} \frac{\mu_j - r_{ft}}{\gamma\sigma^2_j} Prob(S_t = j), 0\right),$$

where μ_j and σ_j are the estimated mean return and standard deviation of the market returns in state j.

The empirical study in this section uses the monthly capital gain and total returns on the S&P 500 index from January 1957 to December 2020. These data are provided by the CRSP. The risk-free rate of return over the same period is the Treasury-bill rate. We simulate the returns to the buy-and-hold strategy, the returns to the strategy based on the assumption that there are 2 states in the market, and the returns to the strategy based on the assumption that there are 5 states in the market. As before, the market regimes are identified using the time series of the capital gain returns. The returns to the regime-based strategies are simulated using an expanding in-sample window procedure. That is, the in-sample period always starts with the first historical observation and ends with

Table 6 Descriptive statistics and performance of the buy-and-hold strategy and the regime-based asset allocation strategies

	Buy-and-hold	HSMM-2	HSMM-5
Performance from 1990 to 2020			
Mean return	11.05	11.05	11.05
Standard deviation	14.64	14.52	10.55
Sharpe ratio	0.58	0.58	**0.80**
P-value		0.49	0.05
Performance from 2005 to 2020			
Mean return	10.41	10.41	10.41
Standard deviation	14.79	13.71	9.70
Sharpe ratio	0.62	0.67	**0.94**
P-value		0.39	0.04

HSMM-2 denotes the strategy based on the 2-state HSMM. *HSMM-5* denotes the strategy based on the 5-state HSMM. The mean returns and standard deviations are annualized and reported in percentages. The Sharpe ratios are also annualized. The *P* values are from the null hypothesis test about the equal performance of the regime-based strategy and the buy-and-hold strategy. Bold values indicate statistical significance at the 5% level

the observation at time t. The optimal asset allocation is determined for time $t + 1$. On each iteration of the procedure, the last observation in the in-sample window is pushed forward by one observation.

We measure the performance by the Sharpe ratio. After the simulations, we test the null hypothesis that the Sharpe ratio of the regime-based strategy equals the Sharpe ratio of the buy-and-hold strategy. The null and alternative hypotheses in our test are

$$H_0 : SR_{HSMM} = SR_{BH}, \quad H_A : SR_{HSMM} > SR_{BH},$$

where SR_{HSMM} and SR_{BH} are the Sharpe ratio of the regime-based and buy-and-hold strategies, respectively. This test is conducted using the Jobson and Korkie (1981) test corrected by Memmel (2003).

The regime-based strategy's Sharpe ratio does not depend on the choice of γ. We select γ to ensure that the mean return of the regime-based strategy equals that of the buy-and-hold strategy. As a result, the mean return is the same in all alternative strategies. This approach allows us not only to evaluate the superiority of the regime-based strategies but also to evaluate their risk reduction properties.

Table 6 reports the descriptive statistics and performance of the buy-and-hold strategy and the regime-based asset allocation strategies for two arbitrary choices of the out-of-sample period. Our main findings can be summarized as follows. Regardless of the choice of the out-of-sample period, the riskiness and performance of the asset allocation strategy based on the 2-state HSMM are virtually the same as in the buy-and-hold strategy. In stark contrast, the strategy based on the five-state HSMM allows the investor to reduce the risk by 30–35% and increase the Sharpe ratio by 35–50%.

The empirical study in this section reinforces the advantages of the 5-state HSMM over the conventional 2-state HSMM. The first advantage is that the 5-state HSMM is better than the 2-state HSMM in regime detection. The second advantage is that the asset allocation in the 5-state HSMM is more optimal than in the 2-state HSMM. Specifically, the 2-state HSMM assumes that the bull market is a high-return low-volatility state, whereas the bear market is a low-return high-volatility state. However, our results suggest that this assumption is basically wrong. Therefore, the 2-state HSMM can easily confuse a high-volatility bull market with a bear market. Additionally, the HSMM-2 strategy is essentially a "bang-bang" strategy that allocates a particular fraction to the stocks if the decoded state is the bull market or zero if the decoded state is the bear market. However, the bang-bang strategy is clearly suboptimal because there are several types of bull and bear market states, and each market state has its own risk-reward pattern.

Summary and conclusions

The dating of bull and bear markets can be done using non-parametric methods based on a set of rules. These methods identify the periods where the mean market return is either positive or negative. However, the market dynamics are much more complex than this dichotomous classification. Multi-state regime-switching Markov models allow one to capture the high degree of complexity of market dynamics.

This paper uses the HSMM and a novel model selection procedure to determine different states in the US stock market. The methodology used in the paper has two motivations. First, virtually all existing studies employ the HMM, where the duration of bull and bear markets does not depend on the market age. The HSMM generalizes the HMM by allowing for duration dependence. Second, existing studies either specify a-priory the number of market states or select the number of states using the standard model selection criteria that do not consider the quality and realism of the competing models. This paper proposes a model selection criterion that evaluates the quality of a model.

The empirical results suggest that the US stock market is switching between five states that can be classified into three bull states and two bear states. The three bull states are categorized as a low-volatility bull market, a high-volatility bull market, and a stock market bubble. One of the bear states represents a regular bear market, while the other one corresponds to either a stock market crash or a market correction. The paper demonstrates that the five-state model is consistent with a number of stylized facts and provides many valuable insights into the regime-switching dynamics of the US stock market and the risk-reward pattern of each regime. Besides, the five-state model reveals the weaknesses of some popular active strategies that aim to enhance returns and reduce losses.

The paper also considers the capital allocation between the stock market and the risk-free asset, where the decision on the optimal stock investment depends on the prevailing market regime. In out-of-sample tests, the asset allocation strategy based on the five-state model achieves higher performance with lower risk than the strategy based on the two-state model and the buy-and-hold benchmark. Whereas the riskiness and performance of the asset allocation strategy based on the conventional two-state HSMM are the same as in the buy-and-hold benchmark, the strategy based on the five-state HSMM allows the investor to reduce the risk by 30–35% and increase the Sharpe ratio by 35–50%. Therefore, the five-state HSMM enables academics and investment professionals not only to understand better the dynamics of the stock market but also to make better asset allocation decisions.

Appendix

See Table 7.

Table 7 Dating of decoded states in the 5-state HSMM for the S&P 500 index

State	Start	End	Duration	State	Start	End	Duration
2	Jan 1957	Feb 1958	14	2	Aug 1989	Jul 1990	12
1	Mar 1958	Jul 1959	17	4	Aug 1990	Sep 1990	2
2	Aug 1959	Oct 1960	15	5	Oct 1990	Mar 1992	18
3	Nov 1960	Feb 1962	16	1	Apr 1992	Oct 1993	19
4	Mar 1962	Jun 1962	4	2	Nov 1993	Nov 1994	13
5	Jul 1962	Sep 1963	15	1	Dec 1994	Jun 1996	19
1	Oct 1963	Feb 1965	17	3	Jul 1996	Jul 1997	13
2	Mar 1965	Jul 1966	17	4	Aug 1997	Oct 1997	3
4	Aug 1966	Sep 1966	2	3	Nov 1997	Jul 1998	9
5	Oct 1966	Jun 1968	21	4	Aug 1998	Oct 1998	3
2	Jul 1968	Mar 1970	21	5	Nov 1998	Aug 2000	22
4	Apr 1970	Jun 1970	3	2	Sep 2000	May 2002	21
5	Jul 1970	Dec 1971	18	4	Jun 2002	Sep 2002	4
1	Jan 1972	Dec 1972	12	5	Oct 2002	Jan 2004	16
2	Jan 1973	Jun 1974	18	2	Feb 2004	Apr 2005	15
4	Jul 1974	Nov 1974	5	1	May 2005	Jan 2007	21
5	Dec 1974	Sep 1976	22	2	Feb 2007	Aug 2008	19
2	Oct 1976	Jun 1978	21	4	Sep 2008	Feb 2009	6
4	Jul 1978	Oct 1978	4	3	Mar 2009	Apr 2010	14
5	Nov 1978	Nov 1980	25	4	May 2010	Aug 2010	4
2	Dec 1980	Jul 1982	20	5	Sep 2010	Jun 2012	22
4	Aug 1982	Oct 1982	3	1	Jul 2012	Jun 2014	24
5	Nov 1982	Jan 1984	15	2	Jul 2014	Mar 2016	21
2	Feb 1984	Sep 1985	20	1	Apr 2016	Nov 2017	20
3	Oct 1985	Jun 1986	9	3	Dec 2017	Sep 2018	10
4	Jul 1986	Sep 1986	3	4	Oct 2018	Dec 2018	3
3	Oct 1986	Aug 1987	11	3	Jan 2019	Jan 2020	13
4	Sep 1987	Nov 1987	3	4	Feb 2020	Apr 2020	3
5	Dec 1987	Jul 1989	20	5	May 2020	Dec 2020	7

Duration is measured in months

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